

# MdMTA-mediated m<sup>6</sup>A modification enhances drought tolerance by promoting mRNA stability and translation efficiency of genes involved in lignin deposition and oxidative stress

Nan Hou<sup>1</sup> , Chaoshuo Li<sup>1</sup> , Jieqiang He<sup>1</sup> , Yu Liu<sup>1</sup> , Sisi Yu<sup>1</sup> , Mickael Malnoy<sup>2</sup> ,  
Muhammad Mobeen Tahir<sup>1</sup> , Lingfei Xu<sup>1</sup> , Fengwang Ma<sup>1</sup>  and Qingmei Guan<sup>1</sup> 

<sup>1</sup>State Key Laboratory of Crop Stress Biology for Arid Areas/Shaanxi Key Laboratory of Apple, College of Horticulture, Northwest A&F University, No. 3 Taicheng Road, Yangling, Shaanxi 712100, China; <sup>2</sup>Research and Innovation Centre, Fondazione Edmund Mach, Via E. Mach, San Michele all'Adige 38010, Italy

Author for correspondence:  
Qingmei Guan  
Email: qguan@nwfau.edu.cn

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## Summary

- Although the N<sup>6</sup>-methyladenosine (m<sup>6</sup>A) modification is the most prevalent RNA modification in eukaryotes, the global m<sup>6</sup>A modification landscape and its molecular regulatory mechanism in response to drought stress remain unclear.
- Transcriptome-wide m<sup>6</sup>A methylome profiling revealed that m<sup>6</sup>A is mainly enriched in the coding sequence and 3' untranslated region in response to drought stress in apple, by recognizing the plant-specific sequence motif UGUAAH (H=A, U or C). We identified a catalytically active component of the m<sup>6</sup>A methyltransferase complex, MdMTA. An *in vitro* methyl transfer assay, dot blot, LC-MS/MS and m<sup>6</sup>A-sequencing (m<sup>6</sup>A-seq) suggested that MdMTA is an m<sup>6</sup>A writer and essential for m<sup>6</sup>A mRNA modification.
- Further studies revealed that MdMTA is required for apple drought tolerance. m<sup>6</sup>A-seq and RNA-seq analyses under drought conditions showed that MdMTA mediates m<sup>6</sup>A modification and transcripts of mRNAs involved in oxidative stress and lignin deposition. Moreover, m<sup>6</sup>A modification promotes mRNA stability and the translation efficiency of these genes in response to drought stress. Consistently, MdMTA enhances lignin deposition and scavenging of reactive oxygen species under drought conditions.
- Our results reveal the global involvement of m<sup>6</sup>A modification in the drought response of perennial apple trees and illustrate its molecular mechanisms, thereby providing candidate genes for the breeding of stress-tolerant apple cultivars.

## Introduction

As a basic component of all living organisms, RNA transmits genetic information from DNA to protein. During this process, various modifications of RNA transcripts have occurred. Since the 1950s, more than 150 modifications have been found in RNA (Boccalletto *et al.*, 2018), among which N<sup>6</sup>-methyladenosine (m<sup>6</sup>A) is the most common modification (Meyer *et al.*, 2012). m<sup>6</sup>A is involved in several mRNA metabolic processes, such as nuclear–cytoplasmic export (Fustin *et al.*, 2013; Roundtree *et al.*, 2017b), RNA stability (X. Wang *et al.*, 2014; Y. Wang *et al.*, 2014; Shi *et al.*, 2017; Huang *et al.*, 2018), pre-mRNA splicing (Zhao *et al.*, 2014; Xiao *et al.*, 2016), primary-microRNA processing (Xiao *et al.*, 2016; Bhat *et al.*, 2020), alternative polyadenylation (APA) site choice (Kasowitz *et al.*, 2018; Yue *et al.*, 2018) and translation efficiency (Meyer *et al.*, 2015; Wang *et al.*, 2015). m<sup>6</sup>A is a dynamic and reversible modification composed of three main components, including methyltransferase-‘writers’, binding protein-‘readers’ and demethylase-‘erasers’ (Saneyoshi *et al.*, 1969; Meyer & Jaffrey,

2017; Roundtree *et al.*, 2017a). Writers are protein complexes that include METTL3, METTL14, WTAP and other functional proteins, such as METTL16, RBM15, ZC3H13, VIRMA and HAKAI (Wei *et al.*, 1975; Ping *et al.*, 2014; Patil *et al.*, 2016; Růžicka *et al.*, 2017; Knuckles *et al.*, 2018; Yue *et al.*, 2018). The m<sup>6</sup>A methyltransferases in plants are highly conserved in animals. In *Arabidopsis thaliana*, the methyltransferases that have been identified are MTA (ortholog of METTL3), MTB (ortholog of METTL14), FIP37 (ortholog of WTAP), VIRILIZER and HAKAI (Zhong *et al.*, 2008; Shen *et al.*, 2016; Růžicka *et al.*, 2017).

In recent years, m<sup>6</sup>A-related research in plants has increased. In *A. thaliana*, m<sup>6</sup>A regulates embryonic development, shoot stem cell fate, floral transition, defective leaf initiation and trichome morphogenesis (Zhong *et al.*, 2008; Shen *et al.*, 2016; Duan *et al.*, 2017; Wei *et al.*, 2018). m<sup>6</sup>A is essential for early sporogenesis in rice and fruit ripening in tomato and strawberry (Zhang *et al.*, 2019; Zhou *et al.*, 2019, 2021), suggesting an important role of m<sup>6</sup>A in plant growth and development. Moreover, an increasing number of studies have shown that m<sup>6</sup>A is

involved in different abiotic stresses. In *Arabidopsis*, m<sup>6</sup>A participates in the salt stress response by regulating the balance of reactive oxygen species (ROS) (Hu *et al.*, 2021). Under salt, drought and heat stress, the *Arabidopsis* mutant *alkbb6* shows a lower survival rate than the wild-type (Huong *et al.*, 2020). The function of m<sup>6</sup>A levels under heat stress has also been investigated in pak-choi (Liu *et al.*, 2020). In poplar, *PtrMTA* overexpression plants have higher trichome density and a more developed root system (Lu *et al.*, 2020). However, current studies have primarily focused on the biological function of m<sup>6</sup>A methylases in response to abiotic stresses, but our understanding of their molecular mechanisms remains very limited.

Drought stress is a major limiting factor for crop production worldwide, and its frequency and intensity are increasing owing to ongoing climate change (Zhou *et al.*, 2017). In perennial woody plants, lignin provides mechanical support and participates in water transport, and it is positively correlated with drought tolerance (Moura-Sobczak *et al.*, 2011; Lee *et al.*, 2012). In rice, overexpression of *OsTFIL* and *OsERF71* can increase plant drought resistance by regulating the expression of lignin synthesis genes (Lee *et al.*, 2016; Bang *et al.*, 2019). In apple, MdMYB88 and MdMYB124 positively regulate lignin synthesis under drought stress by binding to the promoters of *MdMYB46* and *MdVND6*, two important regulators of secondary cell wall biosynthesis (Geng *et al.*, 2018). When exposed to drought stress, ROS and malondialdehyde (MDA) accumulate in plant cells (Pallavi *et al.*, 2012; Liu *et al.*, 2014a,b). Plants have evolved multiple strategies to tolerate oxidative stress that results from ROS accumulation, including the enzymatic and nonenzymatic antioxidant defense systems (Mittler *et al.*, 2004; Gill & Tuteja, 2010; Mittler, 2017). There is also growing evidence that epigenetic regulation, such as DNA methylation and histone modification, plays a role in the transcriptional regulation of genes in response to drought stress (Filichkin *et al.*, 2018), although this has been demonstrated mainly in model plants (Munshi *et al.*, 2015; Mozgova *et al.*, 2019; Godwin & Farrona, 2020). However, only a few studies on the relationship between RNA methylation and drought stress tolerance have been reported in plants. Recent studies in poplar show only the biological function of *PtrMTA* in the drought response (Lu *et al.*, 2020). The transcriptome-wide m<sup>6</sup>A methylome landscape and the specific regulatory mechanism in response to drought remain largely unclear.

In this study, we performed transcriptome-wide m<sup>6</sup>A profiling in apple and investigated changes in m<sup>6</sup>A modification under drought stress. Combined m<sup>6</sup>A-sequencing (m<sup>6</sup>A-seq) and RNA-seq analysis showed that drought altered the m<sup>6</sup>A modification levels of drought-responsive genes, thereby affecting their expression levels. To gain insight into the role of m<sup>6</sup>A in drought stress, we generated transgenic plants of one methyltransferase, MdMTA, which mainly affected the m<sup>6</sup>A levels at 3' untranslated region (UTR) and coding sequence (CDS) regions. MdMTA is a positive regulator of drought tolerance. In addition, m<sup>6</sup>A-seq analysis showed that MdMTA-mediated m<sup>6</sup>A modification positively regulated the mRNA stability and translation efficiency of genes involved in the lignin metabolic process and oxidative stress

under drought conditions. Overall, our work establishes a molecular link between m<sup>6</sup>A mRNA modification and drought tolerance in apple trees.

## Materials and Methods

### Generation of transgenic apple plants

The CDS of *MdMTA* was cloned into the PKGMYC vector to generate the overexpression plasmid, and a 300 bp fragment of the *MdMTA* sequence was cloned into the interference vector pK7WIWG2D. The overexpression or interference plasmid of *MdMTA* was transformed into *Agrobacterium* EHA105, and then transformed into GL-3 as previously reported (Dai *et al.*, 2013). The primers used are listed in Supporting Information Table S1.

### m<sup>6</sup>A-immunoprecipitation (IP)-qPCR

m<sup>6</sup>A-IP-quantitative PCR (qPCR) was performed as previously reported (Dominissini *et al.*, 2013). Briefly, the input RNA and immunoprecipitated RNA were reverse transcribed with random hexamers (R223-01; Vazyme, Nanjing, China). The relative mRNA enrichment was measured using quantitative reverse transcriptase (qRT)-PCR and normalized to the input level. The primers used are listed in Table S1.

### In vitro methylation assay

The coding regions of *MdMTA* and *MdMTB* were amplified by PCR and cloned into the pMAL-c5X vector. The resulting constructs (MBP-*MdMTA* and MBP-*MdMTB*) were transformed into *Escherichia coli* strain Rosetta-gami (DE3). The purification of MBP-*MdMTA* and MBP-*MdMTB* was performed as per the manufacturer's instructions (New England Biolabs, Ipswich, MA, USA). An *in vitro* methylation assay was performed as previously described (Liu *et al.*, 2014a,b). Briefly, a total volume of 50 µl of reaction mixture was used with the following components: 0.15 nmol RNA probe (synthesized in Genscript), 0.15 nmol proteins (MdMTA and MdMTB), 0.8 mM SAM (A4377; Sigma), 80 mM KCl, 1.5 mM MgCl<sub>2</sub>, 0.2 U µl<sup>-1</sup> RNasin, 10 mM dithiothreitol (DTT), 4% glycerol and 15 mM HEPES (pH 7.9). The reaction was incubated at 16°C for 12 h. RNA was extracted twice with chloroform isoamyl alcohol to remove proteins, and then anhydrous ethanol, sodium acetate and glycogen were added overnight at -20°C. After resuspending the RNA with 75% ethanol and anhydrous ethanol, 30 µl of RNase-free water was added to dissolve the RNA. The sequence of the RNA probe was as given by Liu *et al.* (2014a,b). The primers used are listed in Table S1.

### mRNA stability under drought stress

mRNA stability assays were performed according to Zhou *et al.* (2021). Two-month-old plants were cultured hydroponically for 2 wk, and 20% polyethylene glycol (PEG)-8000 was added to the aqueous solution to simulate drought treatment (Geng *et al.*,

2018). Meanwhile, 10  $\mu$ M actinomycin D and dimethyl sulfoxide (DMSO) were added to the plants under PEG treatment and control conditions, respectively. After treatment for 30 min, samples were collected as controls. After 1, 2 and 4 h, samples were taken for downstream gene expression levels. The primers used are listed in Table S1.

### Translation efficiency assay

Translation efficiency assays were performed according to Merchante *et al.* (2015). Briefly, 2 g of apple leaves was used for polysome extraction with an extraction buffer (200 mM Tris-HCl, pH 9.0, 35 mM MgCl<sub>2</sub>, 200 mM KCl, 25 mM EGTA, 1% Triton X-100 (v/v), 1% IGEPAL CA-630 (v/v), 5 mM DTT, 1 mM phenylmethylsulfonyl fluoride (PMSF), 50  $\mu$ g ml<sup>-1</sup> chloramphenicol and 100  $\mu$ g ml<sup>-1</sup> cycloheximide). The samples were added to 8 ml of extraction buffer and placed at 4°C with shaking for 20 min, then centrifuged at 16 000 *g* for 20 min at 4°C. Then, the supernatant was carefully transferred into 8 ml of sucrose buffer (1.75 M sucrose, 400 mM Tris-HCl, pH 9.0, 35 mM MgCl<sub>2</sub>, 5 mM EGTA, 200 mM KCl, 5 mM DTT, 50  $\mu$ g ml<sup>-1</sup> chloramphenicol and 50  $\mu$ g ml<sup>-1</sup> cycloheximide). The supernatant was removed after centrifugation at 200 000 *g* for 4 h at 4°C and the polysomes were collected by adding 300  $\mu$ l of DEPC-treated water at the base. The plant RNA extraction kit (Cat. #DP441; Tiangen) was used to extract total RNAs and polysomal RNA for qRT-PCR analysis. The primers used are listed in Table S1.

## Results

### Transcriptome-wide m<sup>6</sup>A methylome in response to drought stress

To understand whether m<sup>6</sup>A is involved in drought stress, we constructed and sequenced m<sup>6</sup>A-IP and matched input (non-IP control) libraries of apple GL-3 (used as the wild-type) under drought and normal conditions. Each library contained *c.* 30–35 million reads. Other relevant methods are included in Methods S1. Approximately 93–95% clean reads could be aligned to the apple reference genome using HISAT2 v.2.1.0 (Kim *et al.*, 2019) (Table S2). Following data post-processing (BAM-conversion, sorting using SAMTOOLS v.1.9), m<sup>6</sup>A peaks were identified using peak calling and a differential analysis R-package called EXOMEPEAK v.2.16.0 (Kim *et al.*, 2019) with a *P*-value <0.05. Only 'confident peaks' that were detected in three independent biological replicates with high Pearson correlation coefficients in each sample were used for subsequent analysis (Fig. S1). We obtained 12 509 and 13 102 confident peaks (Fig. 1a) under drought and normal conditions, respectively, which encoded 12 088 and 12 898 transcripts. To assess the accuracy of the sequencing results, we randomly selected six m<sup>6</sup>A target genes for validation by m<sup>6</sup>A-IP-qPCR. The m<sup>6</sup>A-IP-qPCR results showed that all of these genes could be verified, implying the reliability of the sequencing results (Fig. S2).

Since multiple regions of a transcript may undergo m<sup>6</sup>A modification (Zhou *et al.*, 2021), we therefore counted the peak

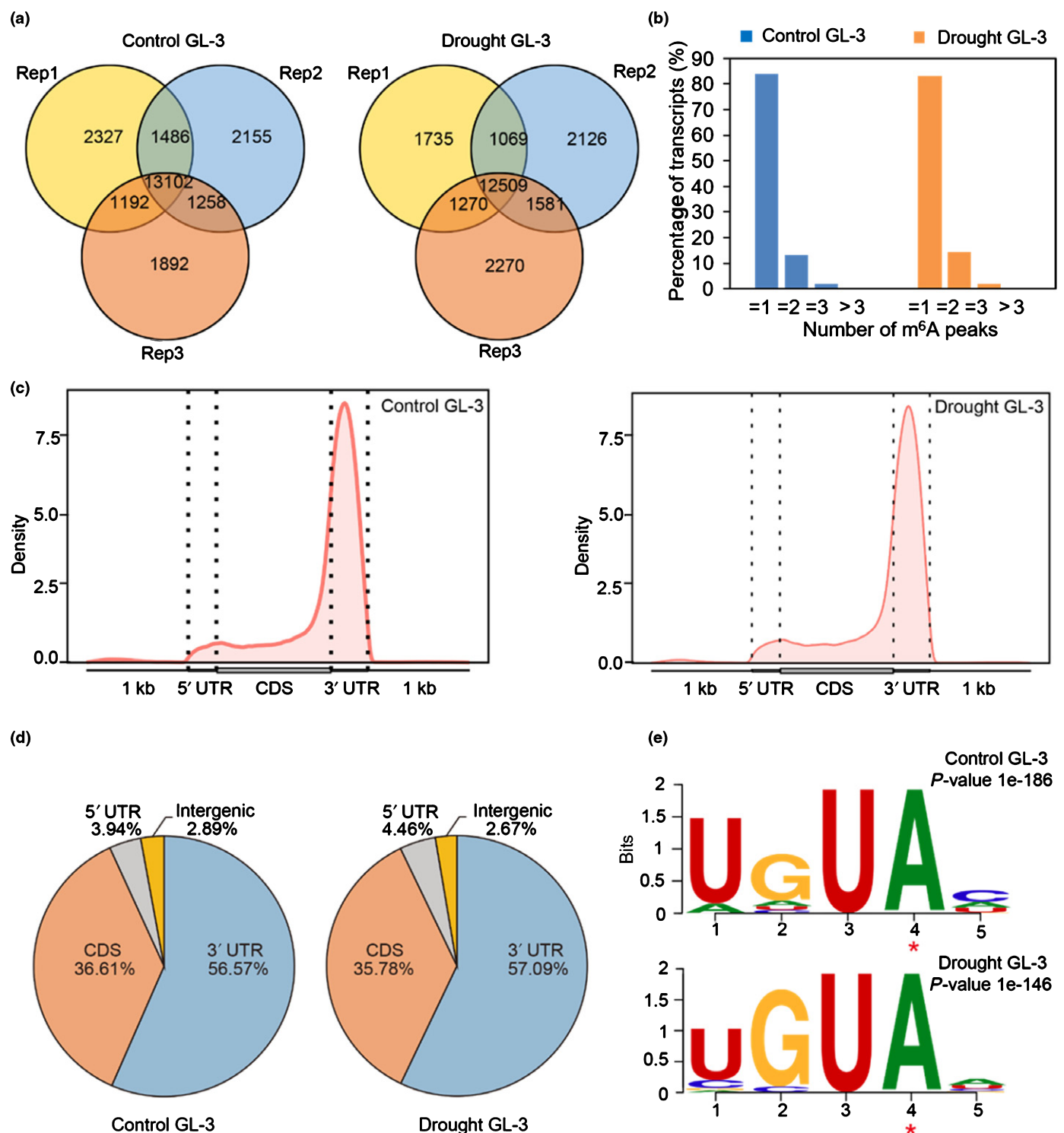
numbers in each transcript. Under both conditions, most transcripts contained only one m<sup>6</sup>A peak, and only a few had two, three or even more peaks (Fig. 1b). To explore the distribution of m<sup>6</sup>A peaks at the whole-transcriptome level in apple, we divided the transcripts into four nonoverlapping regions: 5' UTR, CDS, intergenic region and 3' UTR. We found that the m<sup>6</sup>A peaks were significantly enriched in the 3' UTR under normal and drought conditions, similar to reports in mammals and plants (Fig. 1c,d) (Bodi *et al.*, 2012; Dominissini *et al.*, 2012; Meyer *et al.*, 2012). Interestingly, peaks were also significantly enriched in the CDS region, which differed from reports in other plant species. Next, we used hypergeometric optimization of motif enrichment (HOMER) to analyze the sequence motifs that contained m<sup>6</sup>A peaks in apple. Notably, the plant-specific sequence motif UGUAAH (H=A, U or C) was identified (Figs 1e, S3), similar to UGUAY (Y=A, G, U or C) in *A. thaliana* (Wei *et al.*, 2018) and UGUAMM (M=A or C) in *Zea mays* (Luo *et al.*, 2020).

To further understand the impact of drought on m<sup>6</sup>A methylation, we analyzed the differential peaks (fold change  $\geq 1.5$  and *P* < 0.05) identified under normal and drought conditions. After drought stress treatment, the m<sup>6</sup>A enrichment in 1481 and 676 peaks decreased and increased, respectively (Fig. 2a; Table S3). Gene ontology (GO) enrichment analysis showed that some hypermethylated or hypomethylated genes after drought stress were enriched in drought-related pathways, including water deprivation and oxidative stress (Fig. S4). Moreover, we noted that hypomethylated transcripts were also associated with lignin, flavonoid and fatty acid metabolic processes (Fig. S4a). Our data suggest that the m<sup>6</sup>A levels of some drought-responsive genes were influenced by the drought treatment.

As a widespread regulatory mechanism, m<sup>6</sup>A modification regulates gene expression in various physiological processes (Zaccara *et al.*, 2019). To investigate the association between m<sup>6</sup>A levels and gene expression levels in response to drought stress, we performed an RNA-seq analysis under normal and drought conditions. High Pearson correlation coefficients were observed between three independent biological replicates, indicating ideal repeatability (Fig. S5). We first focused on analyzing expression changes of genes with differential m<sup>6</sup>A levels under drought stress. As shown in Fig. 2(b), the number of genes upregulated by m<sup>6</sup>A modification was greater than that of downregulated genes under drought stress. We also found that expression levels of both hypermethylated and hypomethylated genes increased after drought treatment (Fig. 2c,d). Precise analysis of each transcript showed that more transcripts tended to be upregulated, including both hypomethylated and hypermethylated genes (Fig. 2e). This suggests a complex relationship between m<sup>6</sup>A levels and expression levels. We also noted that the number of differentially expressed genes (DEGs) was much greater than the number of differential peaks, perhaps because some DEGs with m<sup>6</sup>A modification may regulate downstream genes that lack m<sup>6</sup>A modification (Table S4).

### MdMTA is a writer of m<sup>6</sup>A and responsive to drought stress

Because drought treatment led to changes in m<sup>6</sup>A levels, we next were interested in the role of m<sup>6</sup>A in apple drought tolerance.

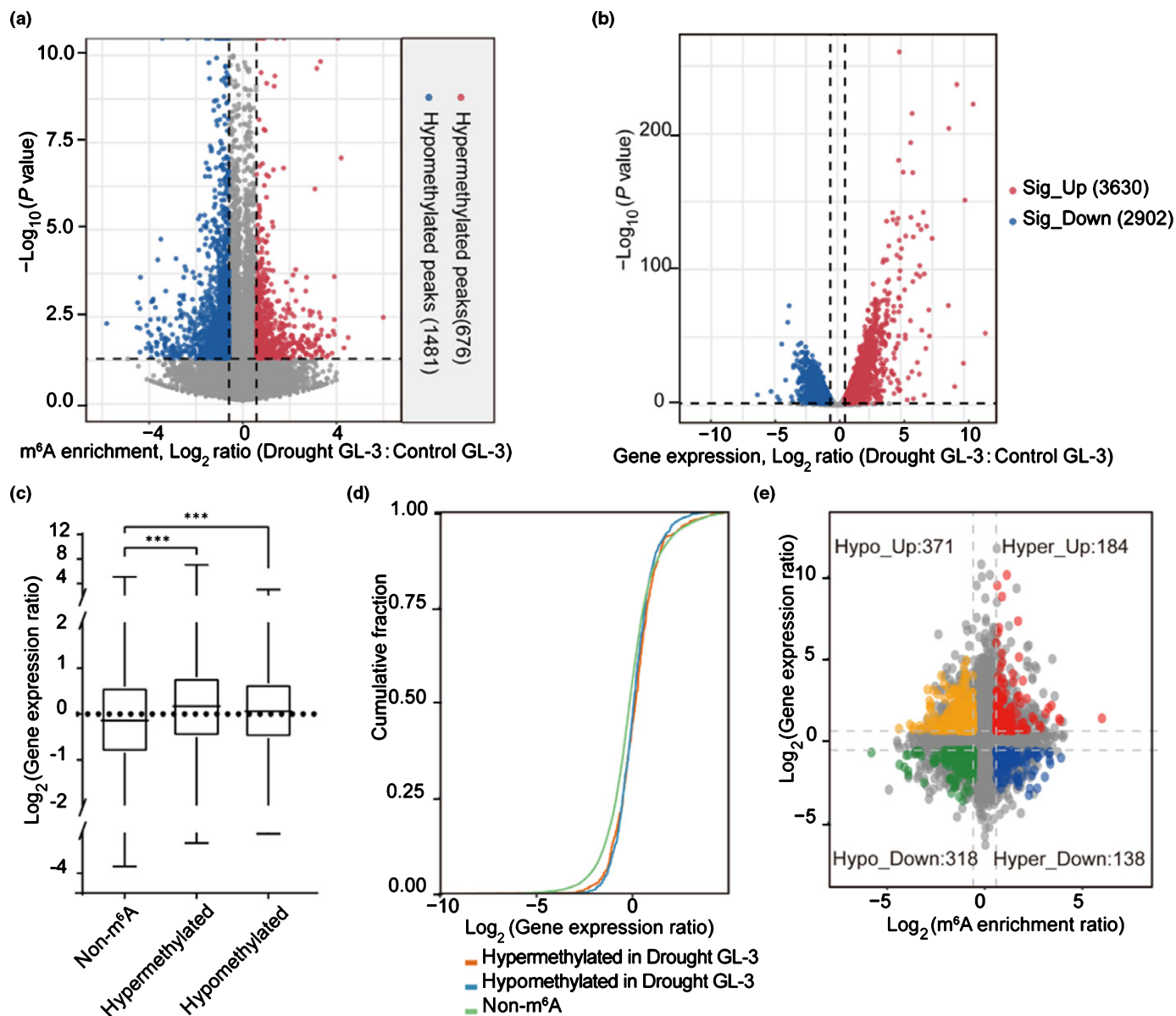


**Fig. 1** Distribution of m<sup>6</sup>A in *Malus domestica* under normal and drought conditions. (a) Venn diagram showing the overlap of three independent m<sup>6</sup>A-seq experiments on GL-3 control and drought treatment groups. Rep, replicate. (b) Percentage of m<sup>6</sup>A-modified transcripts that contained different m<sup>6</sup>A peak numbers among samples. (c) Distribution of m<sup>6</sup>A peaks in transcript segments divided into 5' untranslated region (UTR), coding sequence (CDS) and 3' UTR in GL-3 under normal and drought conditions. The thick box and line represent exons and introns, respectively. (d) Pie chart showing the m<sup>6</sup>A peak distribution. (e) m<sup>6</sup>A-modified sequence motif in GL-3 plants under normal and drought conditions. The red asterisk indicates m<sup>6</sup>A modified adenine.

We then selected the methyltransferase MdMTA for further study. Phylogenetic analysis revealed that MdMTA had a high similarity to *Pyrus bretschneideri* MTA (Fig. 3a).

By transferring the methyl group from SAM to N<sup>6</sup> of adenine, METTL3 completes the installation of m<sup>6</sup>A modification (Bedi *et al.*, 2020). METTL3 (known as MTA in plants) and

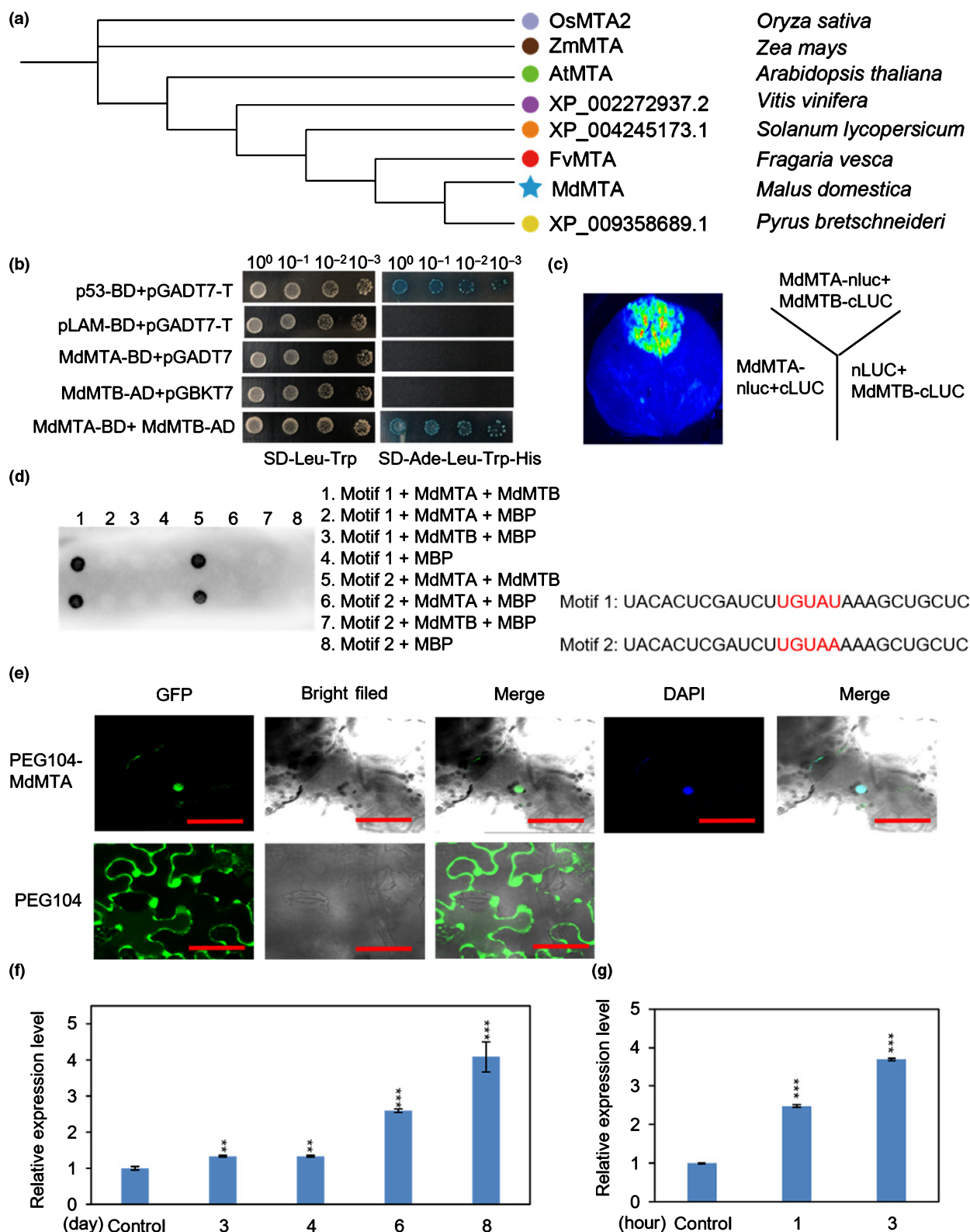




**Fig. 2** Effect of drought on the m<sup>6</sup>A RNA methylome and transcriptome in *Malus domestica*. (a) Volcano plot showing hypermethylated peaks (red) and hypomethylated peaks (blue) under drought treatment compared to normal conditions. The gray plot indicates peaks with no significant difference and the dashed lines denote abline. (b) Volcano plot showing the differential expression of m<sup>6</sup>A-containing genes in GL-3 under drought vs normal conditions. Sig\_Up, significantly upregulated; Sig\_Down, significantly-downregulated. (c) Boxplot comparison of the expression levels of transcript fragments containing m<sup>6</sup>A and non-m<sup>6</sup>A modification under drought vs normal conditions (\*\*\*,  $P < 0.001$ ; Wilcoxon test). These boxplots are composed of boxes showing the 25% and 75% quartiles and the median is represented by the dark line in the center of the box. Their whiskers show the minimum value and maximum value within the boxplot range defined as  $1.5 \times$  the interquartile range. (d) Cumulative distribution of changes in gene expression in (c). (e) Association between m<sup>6</sup>A modification and expression levels of m<sup>6</sup>A-modified genes in GL-3 plants under drought vs normal conditions. Hypo, hypomethylated; Hyper, hypermethylated.

METTL14 (known as MTB in plants) form a stable heterodimer complex that mediates m<sup>6</sup>A deposition (Liu *et al.*, 2014a,b). An *in vitro* methylation assay has shown that only METTL3, not METTL14, has catalytic activity (Wang *et al.*, 2016). In *Arabidopsis*, heterodimerization of MTA and MTB has also been verified (Růžická *et al.*, 2017). We performed a yeast-two hybrid (Y2H) screen using the full-length MdMTA with 778 aa and identified one potential interacting protein, MdMTB (MD07G1276200), which had the highest homology to AtMTB

in *Arabidopsis* (Fig. S6). Y2H analysis (Fig. 3b) and a split-luciferase (Split-LUC) assay (Fig. 3c) confirmed that MdMTA indeed interacted with MdMTB, indicating that MdMTA may possess m<sup>6</sup>A writer characteristics. To verify the methyltransferase ability of MdMTA, we expressed MdMTA and MdMTB proteins in *Escherichia coli* and purified them for an *in vitro* methylation assay (Fig. S7). We found that only in the presence of both MdMTA and MdMTB could the methyl group of SAM be transferred to the adenine of the synthesized RNA probes



**Fig. 3** Identification of the MTA gene in *Malus domestica*. (a) Phylogenetic analysis of MTA in plants. (b) Interaction between MdMTA and MdMTB verified by Y2H assay and (c) Split-Luc assay through the *Nicotiana benthamiana* leaf transient expression system. (d) *In vitro* methylation assay of MdMTA. The red marked part represents the m<sup>6</sup>A motif in apple. (e) Subcellular localization of MdMTA. Bars, 50  $\mu$ m. (f) The expression level of MdMTA in response to drought stress. (g) The expression level of MdMTA in response to polyethylene glycol (PEG) treatment. The asterisks indicate significant differences between the control and treatment groups based on the Tukey's test (\*\*,  $P < 0.01$ ; \*\*\*,  $P < 0.001$ ). The error bars indicate standard deviations ( $n = 3$  in f, g).

(containing a UGUAA/U motif) (Fig. 3d). These results indicated that MdMTA does play a role in methyl transfer in apple.

Subcellular localization results showed that MdMTA was localized in the nucleus (Fig. 3e). To investigate whether *MdMTA* was involved in the drought response, we measured its expression levels after drought and PEG treatments. *MdMTA* was upregulated by drought and PEG treatments with extended treatment time (Fig. 3f,g), indicating that *MdMTA* responds to drought stress.

### MdMTA is essential for m<sup>6</sup>A mRNA modification and plant development

To clarify the function of *MdMTA* in m<sup>6</sup>A modification, we cloned the *MdMTA* gene and performed genetic transformation studies. We obtained RNA interference (RNAi) and overexpression (OE) transgenic lines of *MdMTA* in the GL-3 background (Fig. S8a–c).

To investigate the influence of MdMTA on mRNA methylation, we performed a dot blot and LC-MS/MS assay. The results showed that the m<sup>6</sup>A level of total RNA was lower in *MdMTA* RNAi plants and higher in *MdMTA* OE plants (Fig. S8d,e), which suggested that MdMTA is required for m<sup>6</sup>A modification in apple. To further explore the influence of MdMTA on the global m<sup>6</sup>A methylome, we selected one *MdMTA* RNAi line (#4) that had a lower MdMTA protein level for sequencing analysis. m<sup>6</sup>A methylome analysis identified 12 700 confident peaks with a high Pearson correlation coefficient within 12 552 genes in *MdMTA* RNAi plants (Figs S9, S10a). We found that the m<sup>6</sup>A peak distribution did not differ significantly between *MdMTA* RNAi and GL-3 plants, except for a slight reduction in peaks in the 3' UTR (Figs 4a, S10c,d), consistent with previous work (Yue *et al.*, 2018). *MdMTA* RNAi did not affect the number of m<sup>6</sup>A peaks within each transcript (Fig. S10b). To analyze the effect of *MdMTA* on m<sup>6</sup>A methylation, we investigated the differential peaks between *MdMTA* RNAi and GL-3 plants. Compared with GL-3, there were 630 hypermethylated peaks within 617 genes and 1071 hypomethylated peaks within 1050 genes in *MdMTA* RNAi plants (Fig. 4c,d; Table S5). The hypomethylated peaks were mainly distributed in the 3' UTR and CDS regions, consistent with the m<sup>6</sup>A peak distribution in GL-3. Compared with the m<sup>6</sup>A distribution of GL-3 plants, there was a significant reduction in hypermethylated peaks in the 3' UTR with a concomitant increase in the CDS region in the *MdMTA* RNAi plants (Figs 1d,4b, S10d). GO enrichment analysis indicated that these hypermethylated and hypomethylated genes participated in many biological processes, including RNA processing, RNA metabolism, protein metabolism and histone modification (Fig. S11), consistent with the function of m<sup>6</sup>A. These results suggested that interfering with *MdMTA* reduced the m<sup>6</sup>A levels of most genes with differential peaks, suggesting that MdMTA is essential for m<sup>6</sup>A modification.

To investigate the potential correlation between MdMTA-mediated m<sup>6</sup>A methylation and expression levels, we performed RNA-seq on *MdMTA* RNAi plants using three replicated biological samples (Fig. S12). We identified 2706 upregulated and

2391 downregulated genes (fold change  $\geq 1.5$ ,  $P < 0.05$ ) in *MdMTA* RNAi plants relative to GL-3 (Table S6). Comparison of these DEGs with genes modified by m<sup>6</sup>A methylation revealed that the number of upregulated transcripts (1773) was equal to the number of downregulated transcripts (1628) in *MdMTA* RNAi plants vs GL-3 plants (Fig. 4e–g; Table S7). Based on this observation, we concluded that the effect of MdMTA-mediated m<sup>6</sup>A modification on gene expression was not simply uniform up- or downregulation.

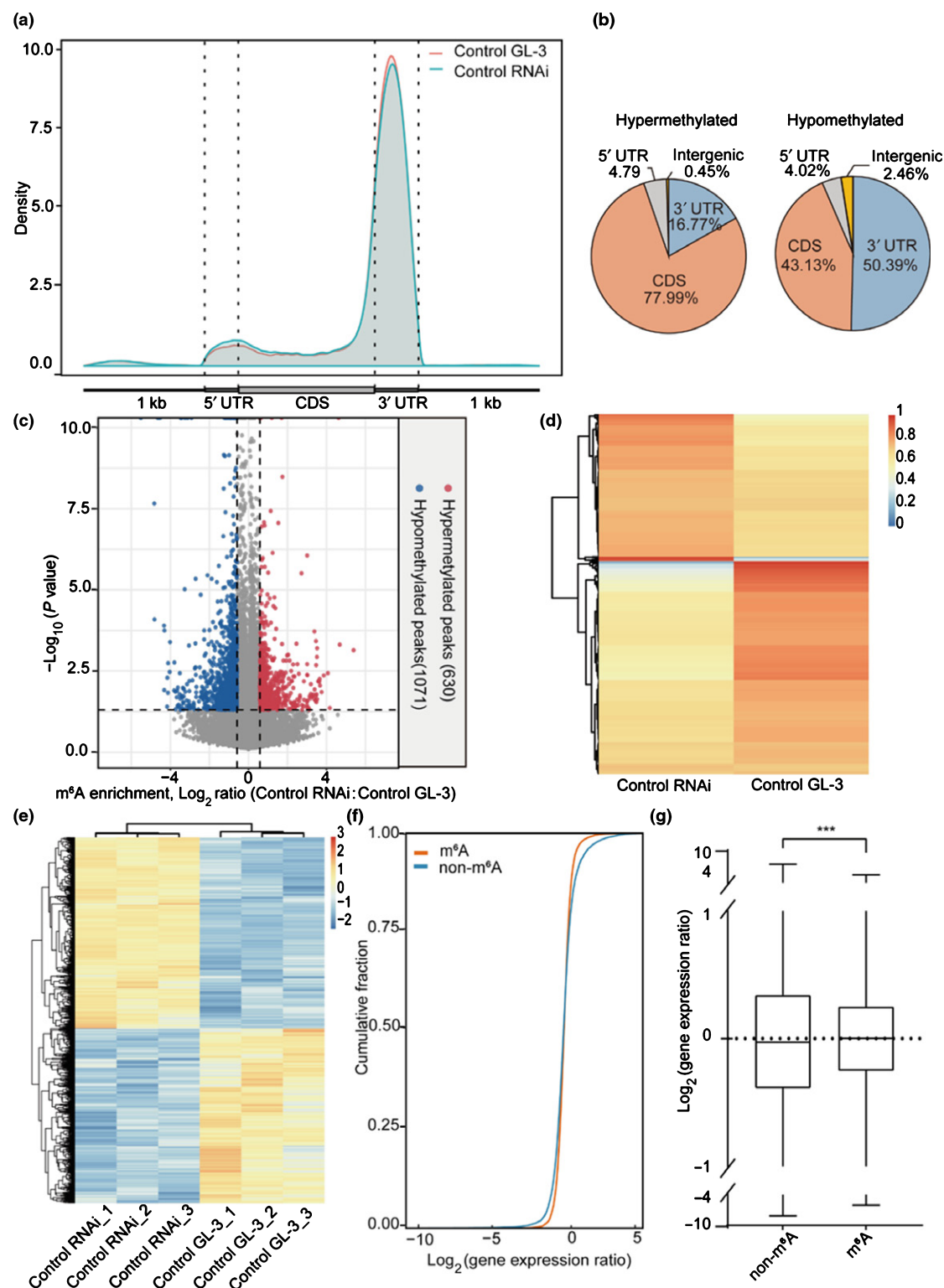
MTA regulates plant development and growth in Arabidopsis and rice (Zhang *et al.*, 2019). To investigate whether MdMTA causes developmental defects in apple, we investigated the morphological characteristics of the 3-month-old *MdMTA* transgenic plants. We found that the plant height of *MdMTA* RNAi plants was lower than that of GL-3 plants, whereas *MdMTA* OE plants had a similar height to GL-3 plants (Fig. 5a). We also found that *MdMTA* RNAi plants had a weaker root system than GL-3 plants (Fig. 5a).

### MdMTA plays a positive role in drought stress tolerance

Because *MdMTA* was induced by drought, we next explored its function in apple drought resistance. To avoid the influence of developmental defects on drought tolerance, we selected the young *MdMTA* transgenic plants and GL-3 with the same growth status for short-term drought treatment. The *MdMTA* RNAi plants showed earlier wilting and lower survival rates than GL-3 plants, whereas the *MdMTA* OE plants showed less wilting and higher survival rates (Figs 5b–d, S13). The electrolyte leakage of *MdMTA* RNAi plants under drought stress was higher than that of GL-3, whereas *MdMTA* OE plants had lower electrolyte leakage (Fig. 5e). Moreover, compared with GL-3, *MdMTA* OE plants had a higher photosynthetic capacity, stomatal conductance, intercellular CO<sub>2</sub> concentration and transpiration rate. By contrast, *MdMTA* RNAi plants had lower levels of these photosynthetic parameters (Fig. S14). These results indicate that *MdMTA* plays a positive regulatory role in drought stress.

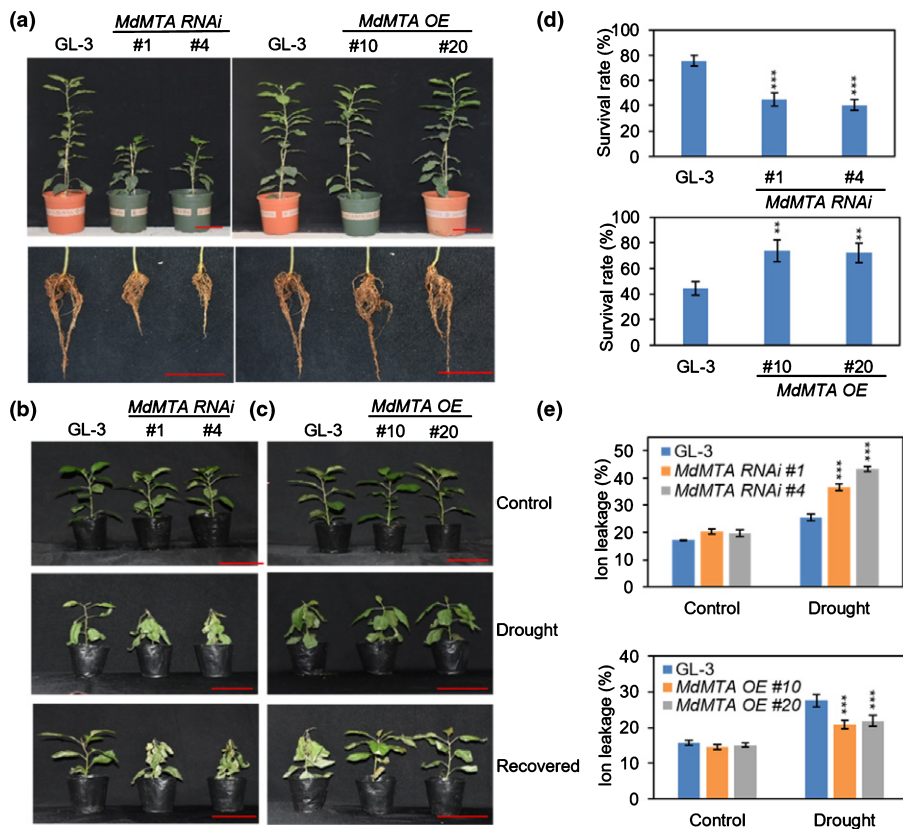
### m<sup>6</sup>A methylation mediated by MdMTA regulates gene expression under drought stress

To gain insight into the molecular mechanism by which *MdMTA* regulates drought tolerance, we performed m<sup>6</sup>A-seq and RNA-seq on *MdMTA* RNAi plants and GL-3 plants after drought treatment. There were three highly replicable independent biological replicates per treatment (Figs S1, S15, S16). Under drought conditions, 11 402 confident peaks were identified in 11 335 genes of the *MdMTA* RNAi plants (Fig. S17). Comprehensive analysis of the m<sup>6</sup>A-seq and RNA-seq data showed that the expression levels of m<sup>6</sup>A-containing transcripts were slightly lower in *MdMTA* RNAi than in GL-3 plants under drought conditions (Fig. 6a). The presence of m<sup>6</sup>A modification has diverse effects on mRNA expression (Luo *et al.*, 2014; Zhou *et al.*, 2021). We subsequently analyzed the expression changes associated with m<sup>6</sup>A modification in different regions of transcripts in *MdMTA* RNAi vs GL-3 plants. As shown in Fig. 6(b), only the expression levels of



**Fig. 4** MdMTA-mediated alterations in the m<sup>6</sup>A RNA methylome and transcriptome. (a) Distribution of m<sup>6</sup>A peaks in GL-3 and *MdMTA* RNAi plants. (b) Distribution of hypermethylated and hypomethylated peaks of *MdMTA* RNAi plants compared with GL-3 plants. (c) Volcano plot showing peaks of hypermethylation (red) and hypomethylation (blue) in *MdMTA* RNAi plants vs GL-3 plants. The gray plot indicates peaks with no significant difference and the dashed lines denote abline. (d) Heatmap showing m<sup>6</sup>A enrichment values with a statistically significant difference in *MdMTA* RNAi plants vs GL-3 plants. (e) Heatmap showing differentially expressed genes with m<sup>6</sup>A modification in *MdMTA* RNAi plants vs GL-3 plants. (f) Cumulative distribution of gene expression changes between *MdMTA* RNAi plants and GL-3 plants for non-m<sup>6</sup>A- and m<sup>6</sup>A-targeted genes. (g) Boxplot comparison of expression levels of non-m<sup>6</sup>A and m<sup>6</sup>A genes in *MdMTA* RNAi plants and GL-3 (\*\*\*,  $P < 0.001$ ; Wilcoxon test). These boxplots are composed of boxes showing the 25% and 75% quartiles and the median is represented by the dark line in the center of the box. Their whiskers show the minimum value and maximum value within the boxplot range defined as  $1.5 \times$  the interquartile range.





**Fig. 5** Drought tolerance of *MdMTA* transgenic plants under short-term drought. (a) Plant height and root phenotype of *MdMTA* transgenic plants under normal conditions. Three-month-old *Malus domestica* plants were grown in a glasshouse for 1 month. Bars, 10 cm. (b, c) Morphological characteristics of GL-3 plants and *MdMTA* transgenic plants with short-term drought stress treatment. Two-month-old plants were treated with drought stress for 24 d (b) and 26 d (c), and then rewatered for 7 d. Bars, 10 cm. (d) Survival rates of plants after the recovery period. (e) Ion leakage of GL-3 plants and *MdMTA* transgenic plants before and after drought treatment. The asterisks indicate significant differences between the GL-3 and transgenic lines based on Tukey's test (\*\*,  $P < 0.01$ ; \*\*\*,  $P < 0.001$ ). The error bars indicate standard deviations ( $n = 10$  in d, 6 in e).

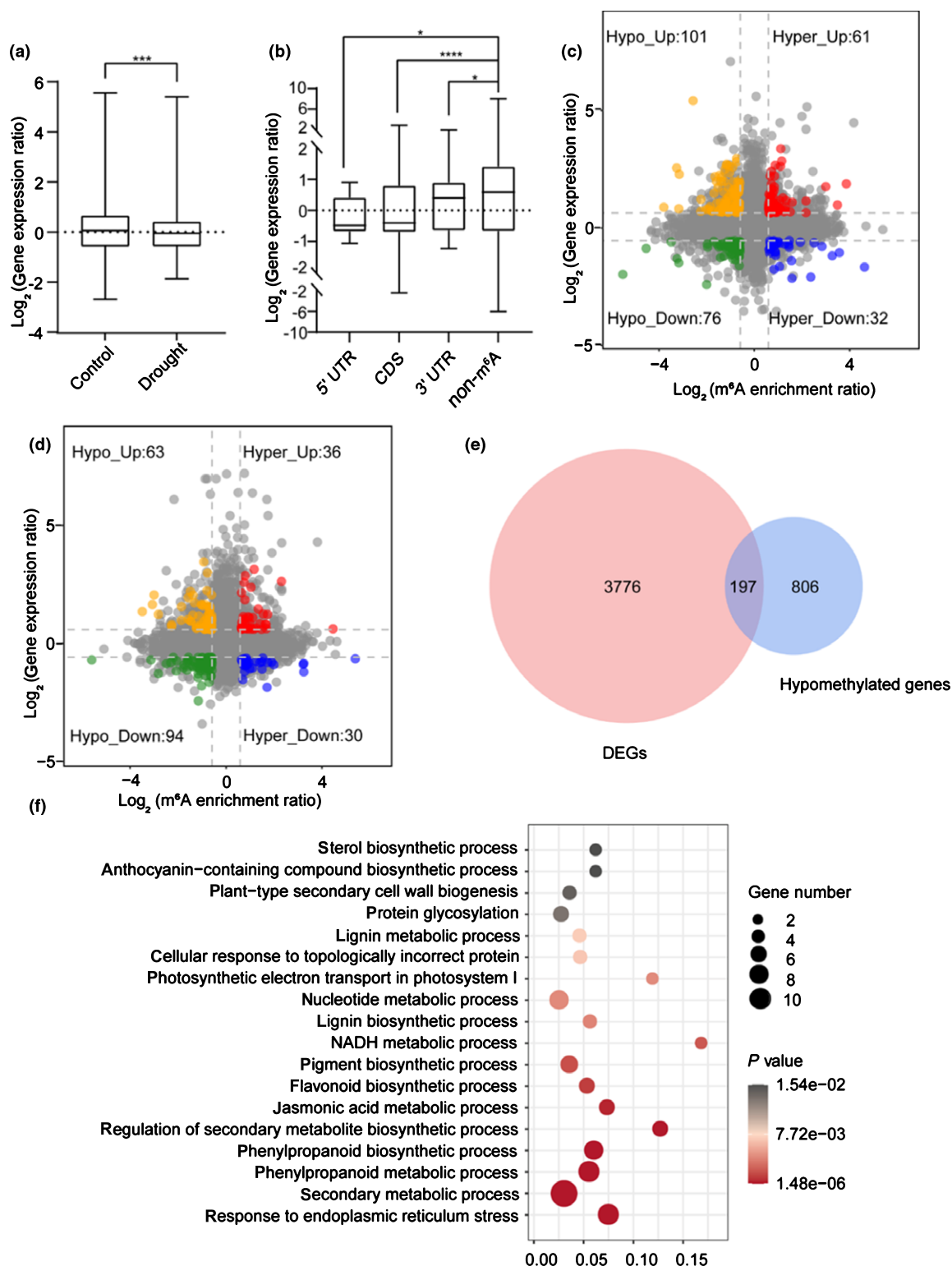
transcripts with m<sup>6</sup>A modification in the 3' UTR were upregulated, whereas transcripts modified in the other regions were downregulated in *MdMTA* RNAi plants. A four-quadrant diagram analysis showed that the expression levels of more genes hypomethylated in *MdMTA* RNAi plants under drought tended to decrease compared with their expression in GL-3 plants, in contrast to the tendency of an increase under normal conditions (Fig. 6c,d).

Subsequently, we focused on the hypomethylated peaks and performed correlation analysis with changes in gene expression levels. Under drought treatment, 197 hypomethylated transcripts were differentially expressed in *MdMTA* RNAi plants (Fig. 6e; Table S8). GO enrichment analysis of these 197 genes revealed that they were enriched in the biosynthetic process of anthocyanin-containing compounds, flavonoid, phenylpropanoid, hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>) and lignin, among others (Fig. 6f). Among the hypomethylated DEGs in *MdMTA* RNAi plants under drought stress, homologs of *MdLEA* are positive regulators of drought stress (Magwanga *et al.*, 2018), homologs of *MdLAC4* and *Md4CL3* are related to lignin biosynthesis (Hahlbrock & Scheel, 1989; Berthet *et al.*, 2011), and homologs of *MdMYC4*, *MdMPV17* and *MdORM2* are related to the H<sub>2</sub>O<sub>2</sub> pathway (Binder *et al.*, 1999; Niu *et al.*, 2011; Li *et al.*, 2016). qRT-PCR analysis showed that the expression of *MdLEA*, *MdLAC4*, *Md4CL3*, *MdORM2*, *MdMYC4* and *MdMPV17* was downregulated in *MdMTA* RNAi plants and upregulated in *MdMTA* OE plants (Fig. 7a), suggesting that *MdMTA* may respond to drought by affecting the expression of genes involved in lignin biosynthesis and H<sub>2</sub>O<sub>2</sub> scavenging under drought.

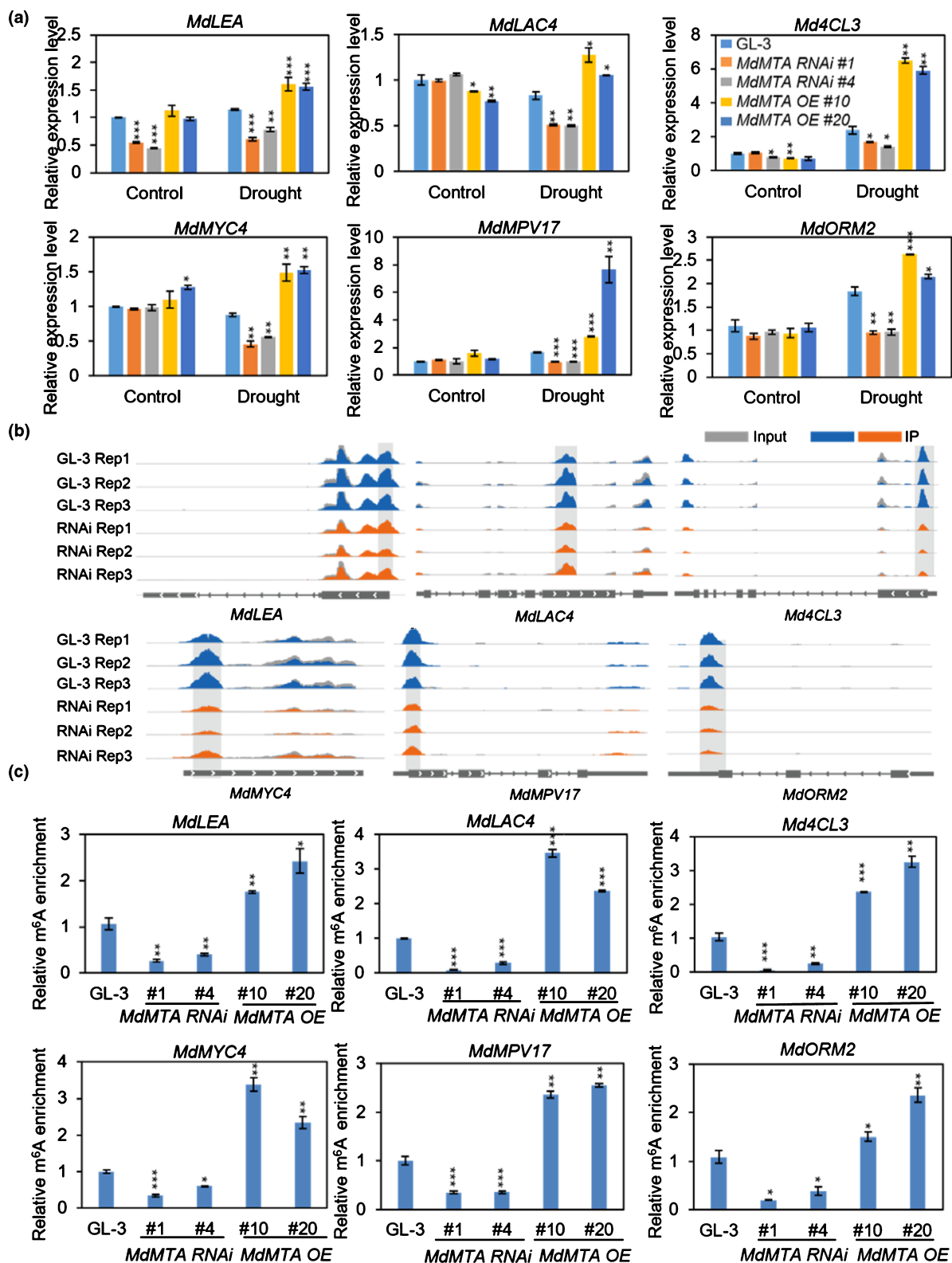
Next, we were interested in whether the *MdMTA*-mediated expression changes in these genes were correlated with m<sup>6</sup>A methylation. We examined the m<sup>6</sup>A levels of these six genes in *MdMTA* transgenic plants using m<sup>6</sup>A-IP-qPCR. The results showed that their m<sup>6</sup>A levels were significantly reduced in the *MdMTA* RNAi plants and increased in the OE plants in response to drought (Fig. 7c), whereas there was no significant difference under normal conditions except for *MdLEA* (Fig. S18b), which is consistent with our m<sup>6</sup>A-seq results (Figs 7b, S18a). These results indicate that *MdMTA*-mediated m<sup>6</sup>A is positively correlated with the expression of *MdLEA*, *MdLAC4*, *Md4CL3*, *MdORM2*, *MdMYC4* and *MdMPV17* under drought stress.

### *MdMTA*-mediated methylation stabilizes mRNAs and enhanced the translation efficiency of mRNAs under drought stress

We then investigated how m<sup>6</sup>A modification affects the transcription level of these six genes under drought stress. Previous studies have shown that m<sup>6</sup>A methylation may be negatively or positively correlated with mRNA stability and gene expression in animals and plants (X. Wang *et al.*, 2014; Y. Wang *et al.*, 2014). To investigate the effect of *MdMTA*-induced m<sup>6</sup>A modification on mRNA stability, we examined the expression levels of these genes under drought conditions after the addition of the transcriptional repressor actinomycin D. The results showed that the stability of *MdLEA*, *MdLAC4*, *MdORM2*, *MdMYC4* and *MdMPV17* mRNA was lower in *MdMTA* RNAi plants than in GL-3 plants under drought stress (Fig. 8a), suggesting that *MdMTA*-

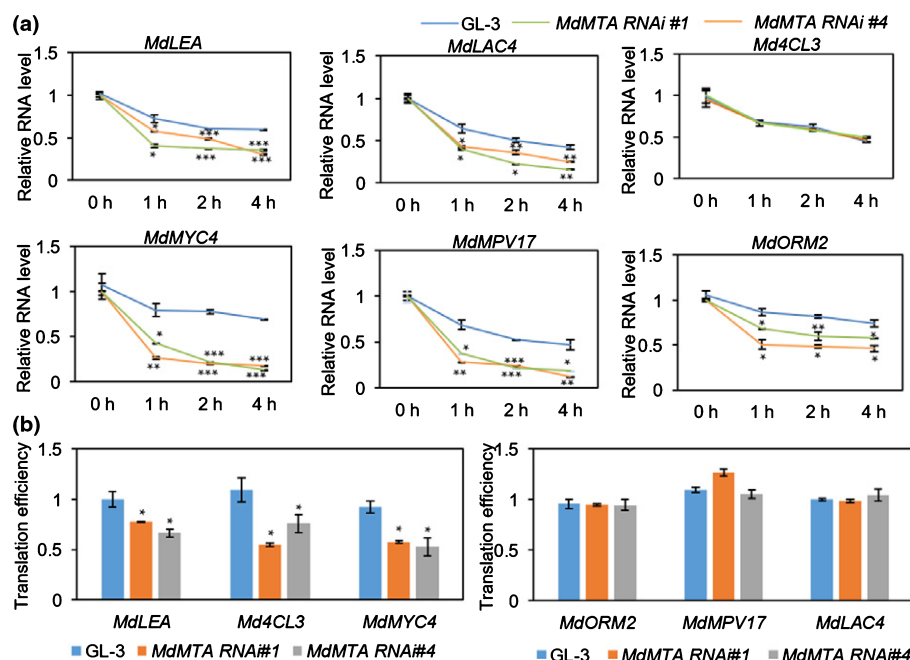


**Fig. 6** *MdMTA*-mediated m<sup>6</sup>A methylation in *Malus domestica* under drought stress. (a) Boxplot comparison of expression levels of m<sup>6</sup>A-modified genes between *MdMTA* RNAi plants and GL-3 plants under control and drought conditions. Two-month-old plants with drought treatment for 20 d were used for m<sup>6</sup>A-seq and RNA-seq. (b) Boxplot comparison of expression levels of non-m<sup>6</sup>A genes and m<sup>6</sup>A genes in different transcript segments between *MdMTA* RNAi plants and GL-3 plants under drought conditions. These boxplots are composed of boxes showing the 25% and 75% quartiles and the median is represented by the dark line in the center of the box. Their whiskers show the minimum value and maximum value within the boxplot range defined as 1.5 × the interquartile range (\*,  $P < 0.05$ ; \*\*\*,  $P < 0.001$ ; Wilcoxon test). (c) Association between m<sup>6</sup>A modification and expression levels of m<sup>6</sup>A-modified genes in GL-3 and *MdMTA* RNAi plants under normal and (d) drought conditions. Hypo, hypomethylated; Hyper, hypermethylated. (e) Venn diagram showing the intersection of hypomethylated genes and differentially expressed genes (DEGs) in *MdMTA* RNAi plants compared with GL-3 plants under drought conditions. (f) Gene Ontology (GO) enrichment of intersection genes from the Venn diagram.



**Fig. 7** MdMTA-mediated m<sup>6</sup>A modification upregulates gene expression level in apple (*Malus domestica*). (a) Transcript levels of *MdLEA*, *MdLAC4*, *Md4CL3*, *MdMYC4*, *MdMPV17* and *MdORM2* determined by qRT-PCR in GL-3 and *MdMTA* transgenic plants under normal and drought conditions. Two-month-old plants with drought treatment for 20 d were used for qRT-PCR. (b) Integrated Genome Viewer displaying the m<sup>6</sup>A-seq read distributions in *MdLEA*, *MdLAC4*, *Md4CL3*, *MdMYC4*, *MdMPV17* and *MdORM2* in GL-3 and *MdMTA* RNAi plants under drought conditions. (c) Validation of the m<sup>6</sup>A enrichment by m<sup>6</sup>A-IP-qPCR. The asterisks indicate significant differences between the GL-3 and transgenic lines based on Tukey's test (\*,  $P < 0.05$ ; \*\*,  $P < 0.01$ ; \*\*\*,  $P < 0.001$ ). The error bars indicate standard deviations ( $n = 3$ ).

**Fig. 8** Reduced m<sup>6</sup>A levels lead to a decrease in mRNA stability and translation efficiency under drought conditions. (a) The degradation rate of the transcripts in GL-3 and *MdMTA* RNAi plants under drought conditions. Six-week-old *Malus domestica* plants were treated with 20% polyethylene glycol (PEG)-8000 and 10  $\mu$ M actinomycin-D for 0, 1, 2 and 4 h. Samples at different time periods were collected for RNA extraction. (b) Translation efficiency of the transcripts in GL-3 and *MdMTA* RNAi plants under drought conditions. Total RNAs and polysomal RNA were extracted from 6-wk-old plants with drought treatment. The abundance ratio of mRNA in the polysomal RNA vs the total RNA was the translation efficiency. The asterisks indicate significant differences between the GL-3 and transgenic lines based on Tukey's test (\*,  $P < 0.05$ ; \*\*,  $P < 0.01$ ; \*\*\*,  $P < 0.001$ ). The error bars indicate standard deviations ( $n = 3$  in a, b).



mediated m<sup>6</sup>A methylation of these genes slows the decay rate of these mRNAs, thereby increasing their expression in response to drought stress. Under normal conditions, the mRNA stability of these transcripts had no difference in *MdMTA* RNAi plants compared with GL-3 plants (Fig. S19a). However, we noted that the stability of *Md4CL3* in *MdMTA* RNAi plants was similar to that of GL-3 plants under drought conditions. m<sup>6</sup>A modification in the 5' UTR can directly bind to eukaryotic initiation factor 3 (eIF3) to promote translation in a cap-independent manner (Meyer *et al.*, 2015). Therefore, we investigated whether *MdMTA* regulated the translation efficiency of *Md4CL3*, which had m<sup>6</sup>A modification in its 5' UTR. We found that the translation efficiency of *Md4CL3* mRNA indeed decreased in *MdMTA* RNAi plants compared to that of GL-3 plants under drought stress (Fig. 8b). Besides, the m<sup>6</sup>A modification in CDS positively regulates translation efficiency by reducing ribosomal pausing (Mao *et al.*, 2019). We found that the translation efficiency of mRNAs modified by the m<sup>6</sup>A modification in CDS had no difference in *MdMTA* RNAi plants in response to drought stress, except for *MdLEA* and *MdMYC4* mRNAs (Fig. 8b). Under normal conditions, the translation efficiency of these genes in *MdMTA* RNAi plants was similar to that in GL-3 plants (Fig. S19b). Together, these results suggest that *MdMTA* responds to drought stress by regulating the mRNA stability and translation efficiency of these genes.

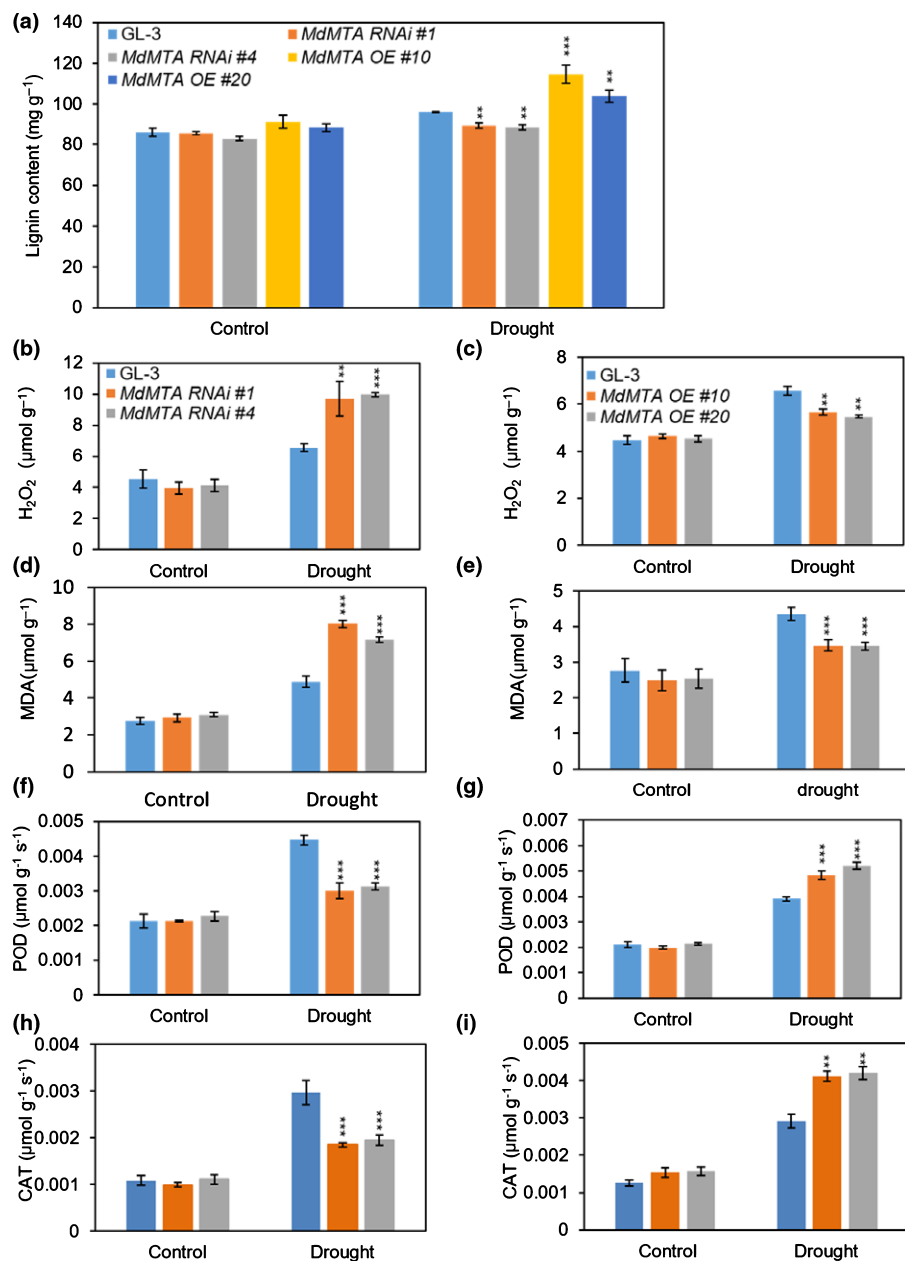
In mouse, knockout of *METTL3* mediates alternative splicing to generate abnormal mRNAs with exon skipping and intron retention (Geula *et al.*, 2015). We next analyzed alternative splicing events based on our RNA-seq. The results revealed a significant difference in exon skipping and mutually exclusive exon events between *MdMTA* RNAi and GL-3 plants (Table S9). However, the alternative splicing patterns of these six genes in *MdMTA* RNAi plants were similar to those in GL-3 plants under drought conditions (Table S10).

*MdMTA* plays a positive role in promoting lignin deposition and scavenging excessive H<sub>2</sub>O<sub>2</sub> under drought stress

*MdMTA* positively regulated m<sup>6</sup>A modification and expression of *MdLEA*, *MdLAC4*, *Md4CL3*, *MdORM2*, *MdMYC4* and *MdMPV17* (Fig. 7). In Arabidopsis, LAC4 with laccase activity is highly expressed in the vascular bundles and interfascicular fibers, and the *lac4* mutant shows a reduced lignin content (Berthet *et al.*, 2011). In the phenylpropanoid pathway, 4-coumaric acid: CoA ligase (4CL) is the enzyme for lignin biosynthesis (Hahlbrock & Scheel, 1989). We measured the lignin content in GL-3 and *MdMTA* transgenic plants. Lignin content was lower in *MdMTA* RNAi plants and higher in *MdMTA* OE plants than in GL-3 plants under drought conditions (Fig. 9a). To further confirm this result, we performed a long-term drought experiment for 2 months. After 2 months of treatment, the lignin content was significantly lower in *MdMTA* RNAi plants than in GL-3 plants (Fig. S20a). In contrast to the short-term drought results, the lignin content of *MdMTA* RNAi plants was also significantly reduced under normal conditions. A decreased lignin content has been shown to affect the hydraulic conductivity of *Populus* (Coleman *et al.*, 2008; Voelker *et al.*, 2011). We found that root hydraulic conductivity was consistently lower in *MdMTA* RNAi plants than in GL-3 plants under normal and drought conditions (Fig. S20b). There was no significant difference in shoot hydraulic conductivity between *MdMTA* RNAi and GL-3 plants under normal conditions; however, under drought conditions, *MdMTA* RNAi plants had lower shoot hydraulic conductivity (Fig. S19c). These data suggest that the effect of *MdMTA* on lignin deposition, at least in part, contributes to its role in apple drought tolerance.

In Arabidopsis, *orm1* amiR (artificial microRNA)-*ORM2* lines accumulate more H<sub>2</sub>O<sub>2</sub> than the wild-type (Li *et al.*, 2016), and overexpression of *MYC4* increased the accumulation of

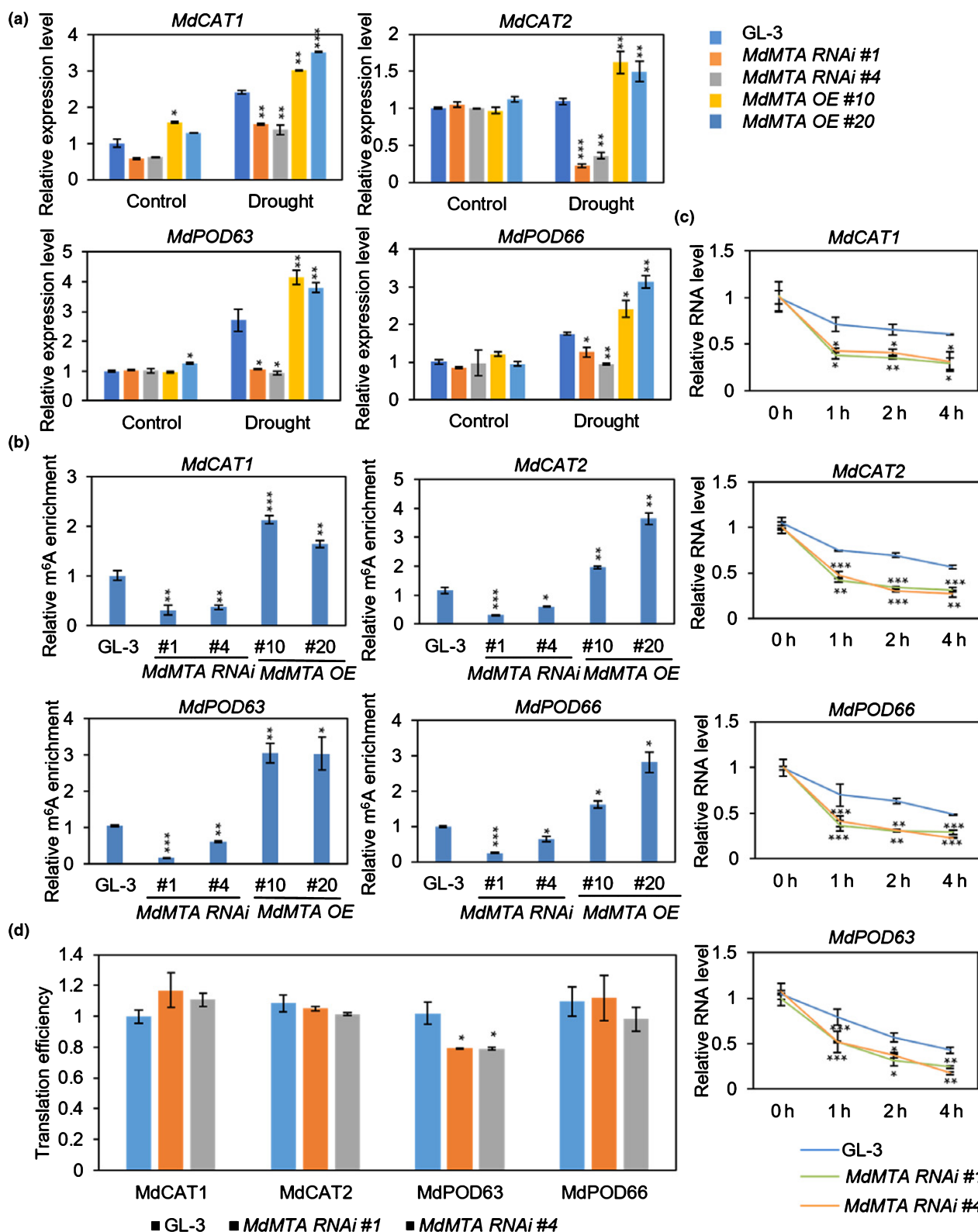




**Fig. 9** Lignin content, hydrogen peroxide and enzyme activity in GL-3 and *MdMTA* transgenic plants under short-term drought treatment. (a) The lignin content in GL-3 and *MdMTA* transgenic plants with drought treatment for 20 d. (b, c) Hydrogen peroxide content. (d, e) MDA activity. (f, g) Peroxidase (POD) activity. (h, i) Catalase (CAT) activity. The asterisks indicate significant differences between the GL-3 and transgenic lines based on Tukey's test (\*\*,  $P < 0.01$ ; \*\*\*,  $P < 0.001$ ). The error bars indicate standard deviations ( $n = 9$  in a–i).

anthocyanin, a nonenzymatic antioxidant (Niu *et al.*, 2011). Loss of *MPV17* function increased ROS content, highlighting its role in peroxisomal reactive oxygen metabolism (Binder *et al.*, 1999; Gottesberge & Felix, 2005; Antonenkov *et al.*, 2015). Therefore, we measured the  $H_2O_2$  content in GL-3 and *MdMTA* transgenic plants. After drought treatment, the  $H_2O_2$  content was higher in *MdMTA* RNAi plants but lower in OE plants, as compared with GL-3 plants (Fig. 9b,c). Under osmotic stress, overexpression of *PyMPV17* reduces the content of MDA (Wi *et al.*, 2020). MDA is one of the most important products of membrane lipid peroxidation, and its content indirectly reflects the degree of damage to the membrane system (Sanocka & Kurpisz, 2004). MDA content was higher in *MdMTA* RNAi plants and lower in *MdMTA* OE plants, as compared with GL-3 under drought stress (Fig. 9d,e), further indicating that the antioxidant defense mechanism was severely

impaired in *MdMTA* RNAi plants. We also examined the activities of the main enzymes in the antioxidant defense system, including catalase (CAT) and peroxidase (POD) (Dinakar *et al.*, 2008). These enzyme activities were significantly lower in *MdMTA* RNAi plants and higher in *MdMTA* OE plants under drought conditions compared with GL-3 (Fig. 9f–i). To explain how *MdMTA* affects the activity of these enzymes, we selected several enzyme-encoded genes with altered expression levels based on RNA-seq of *MdMTA* RNAi and GL-3 plants after drought treatment for verification. The expression level of *MdCAT1*, *MdCAT2*, *MdPOD63* and *MdPOD66* was downregulated in *MdMTA* RNAi plants but upregulated in *MdMTA* OE plants under drought conditions (Fig. 10a). To explore whether the altered expression levels of these genes were due to *MdMTA*-mediated m<sup>6</sup>A modification, we examined the m<sup>6</sup>A levels of these genes in our m<sup>6</sup>A-seq results



**Fig. 10** Reduced m<sup>6</sup>A levels lead to a decrease in mRNA stability and translation efficiency of mRNAs related to CAT and POD enzymes under drought conditions. (a) Transcript levels of *MdCAT1*, *MdCAT2*, *MdPOD63* and *MdPOD66* determined by qRT-PCR in GL-3 and *MdMTA* transgenic plants under normal and drought conditions. Two-month-old plants with drought treatment for 20 d were used for qRT-PCR. (b) Validation of the m<sup>6</sup>A enrichment by m<sup>6</sup>A-IP-qPCR in GL-3 and *MdMTA* transgenic plants under drought conditions. (c) The degradation rate of the transcripts in GL-3 and *MdMTA* RNAi plants under drought conditions. (d) Translation efficiency of the transcripts in GL-3 and *MdMTA* RNAi plants under drought conditions. The asterisks indicate significant differences between the GL-3 and transgenic lines based on Tukey's test (\*,  $P < 0.05$ ; \*\*,  $P < 0.01$ ; \*\*\*,  $P < 0.001$ ). The error bars indicate standard deviations ( $n = 3$  in a–d).

(Table S7). However, the m<sup>6</sup>A modification of these mRNAs was not detected in our m<sup>6</sup>A-seq datasets. We speculated that the limitations of sequencing may have caused this undetectable m<sup>6</sup>A modification of these enzymes. Based on our previously published m<sup>6</sup>A-seq datasets (Mao *et al.*, 2021), we found these genes can be modified by m<sup>6</sup>A modification. Therefore, we designed primers based on where m<sup>6</sup>A modification occur in previous results to detect m<sup>6</sup>A levels of these enzymes in *MdMTA* transgenic plants. m<sup>6</sup>A-IP-qPCR results showed that the m<sup>6</sup>A levels of these genes were lower in *MdMTA* RNAi plants and higher in *MdMTA* OE plants than in GL-3 plants under drought conditions (Fig. 10b). Under normal conditions, the m<sup>6</sup>A levels of these genes showed no difference in *MdMTA* transgenic plants compared with GL-3 plants (Fig. S21a). We also detected the mRNA stability and translation efficiency of these genes. We found that the mRNA stability of these genes was lower than GL-3 plants under drought treatment, but there was no difference under normal conditions (Figs 10c, S21b). Only the translation efficiency of *MdPOD63* with m<sup>6</sup>A modification in the 5' UTR was lower in *MdMTA* RNAi plants than that of GL-3 plants, while the translation efficiency of other genes with m<sup>6</sup>A modification in the 3' UTR in *MdMTA* RNAi plants was similar to that of GL-3 plants under drought conditions (Fig. 10d). There was no difference in translation efficiency of these mRNAs between *MdMTA* RNAi plants and GL-3 plants under normal conditions (Fig. S21c). In addition, the alternative splicing patterns of these genes in *MdMTA* RNAi plants had no difference from GL-3 plants under drought conditions (Table S10). The above results suggest that MdMTA mediated the m<sup>6</sup>A modification of the genes encoding CAT and POD enzymes, thereby regulating the mRNA stability and translation efficiency of these enzymes to promote H<sub>2</sub>O<sub>2</sub> scavenging.

## Discussion

Due to their sessile nature, plants must adapt to biotic and abiotic environmental stresses by perceiving stress factors and changing their molecular characteristics (Zhang *et al.*, 2018). Epigenetic mechanisms constitute a complex regulatory system that helps to control plant responses to severe environmental conditions (Zheng *et al.*, 2017). However, current studies on m<sup>6</sup>A modification have focused mainly on its role in plant development, and only a few studies have addressed its function in stress responses. In addition, the molecular mechanisms by which m<sup>6</sup>A modification responds to drought stress remain unclear. This study found that m<sup>6</sup>A is a common RNA modification in apple and responds to drought stress. Specifically, MdMTA, a methyltransferase that performs m<sup>6</sup>A modification in apple, plays a positive role in drought tolerance by enhancing the mRNA stability and translation efficiency of the genes involved in lignin deposition and ROS scavenging.

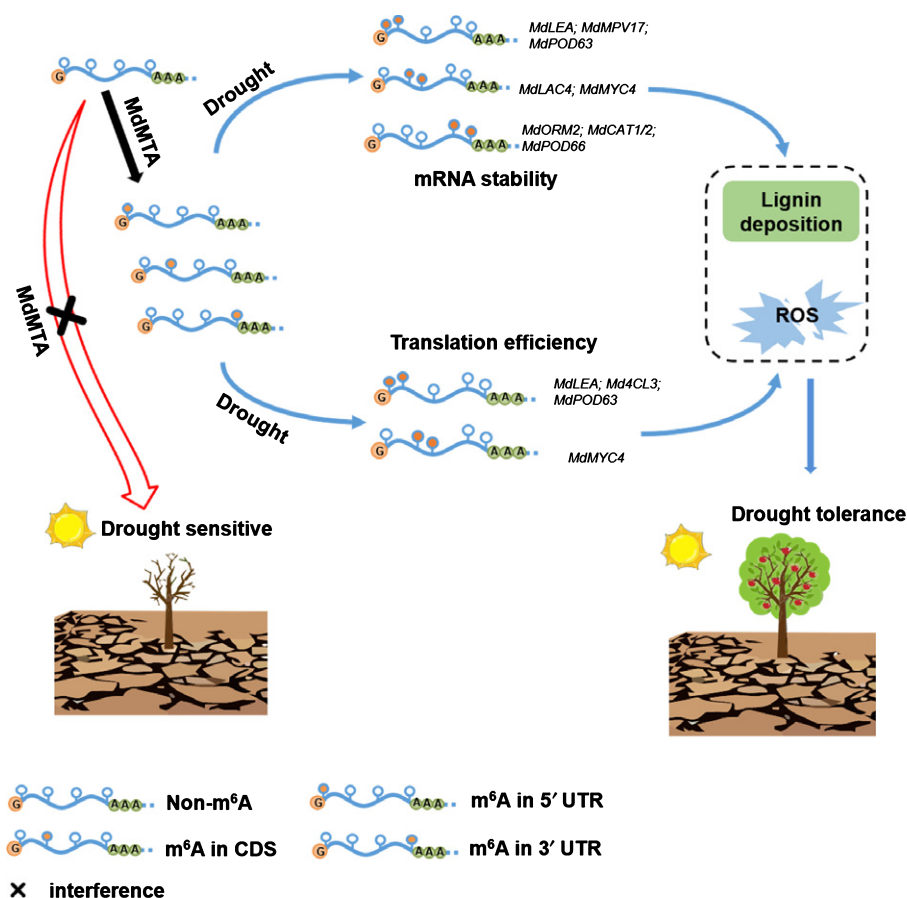
We found that drought cannot change the m<sup>6</sup>A distribution of the 3' UTR and CDS region (Fig. 1c), although it can affect the m<sup>6</sup>A level of drought-responsive genes as shown in Fig. S4. In fact, most of these drought-responsive genes also had the m<sup>6</sup>A modification under normal conditions. Therefore, drought treatment can only increase or decrease their m<sup>6</sup>A level, instead of

affecting their m<sup>6</sup>A pattern. The abiotic stresses reported so far, including salt stress and cold stress, also do not affect the distribution of m<sup>6</sup>A (Hu *et al.*, 2021; Yang *et al.*, 2021).

The m<sup>6</sup>A modification is installed by RNA methyltransferases, including MTA (Yue *et al.*, 2019). In this study, we found that MdMTA was responsible for the installation of the m<sup>6</sup>A modification (Fig. 3d). Several lines of evidence support the indispensable role of MdMTA in m<sup>6</sup>A mRNA modification. First, an *in vitro* methylation assay suggested that MdMTA can transfer the methyl group of SAM to the adenine of the synthesized RNA probes (Fig. 3d). Second, dot blot and LC-MS/MS assays showed that MdMTA positively regulated m<sup>6</sup>A levels (Fig. S7d,e). Third, interference of *MdMTA* expression resulted in a decrease in m<sup>6</sup>A levels at the 3' UTR and CDS regions (Fig. 4b). Fourth, transcriptome-wide m<sup>6</sup>A methylome analysis revealed that there were more hypomethylated transcripts than hypermethylated transcripts in *MdMTA* RNAi plants (Fig. 4c,d). The data presented above indicate that MdMTA plays a vital role in m<sup>6</sup>A installation. We noted that the impact of MdMTA on the distribution of the m<sup>6</sup>A peaks along the transcripts was weaker than that of FIP37 in Arabidopsis (Fig. 4a) (Shen *et al.*, 2016). It may be that knocking out any enzyme of the methyltransferase complex reduces m<sup>6</sup>A installation evenly across most mRNA regions (Yue *et al.*, 2018).

Interfering with *MdMTA* expression did not result in a large number of hypomethylated peaks (Fig. 4c). The reasons for this might be as follows. First, we used interference plants instead of knockout plants. Previous studies have shown that MTA is essential for embryonic development, and knockout *MTA* results in a lethal phenotype in *A. thaliana* (Bodi *et al.*, 2012). We also generated one line of edited plants, but the plants could not be propagated. Therefore, we used the interfering plants for further analysis, which might influence the effect of MdMTA on m<sup>6</sup>A modification. Second, to eliminate duplication bias introduced by library preparation and sequencing, we used Unique Identifier (UMI) sequences in m<sup>6</sup>A-seq and RNA-seq. Therefore, differences that were amplified by PCR were removed, which may lead to a decrease in the number of different peaks or genes. Third, we searched the literature and surveyed the effect of MTA on the global m<sup>6</sup>A methylome in other species. In strawberry, there are 2520 hypomethylated m<sup>6</sup>A peaks covering 1128 gene transcripts and 1510 hypermethylated m<sup>6</sup>A peaks within 1239 gene transcripts in the *MTA* RNAi fruits compared to the control (Zhou *et al.*, 2021). Compared with GL-3, there were 630 hypermethylated peaks within 617 genes and 1071 hypomethylated peaks within 1050 transcripts in *MdMTA* RNAi plants. The number of hypomethylated transcripts (1050) in *MdMTA* RNAi plants was comparable to that (1128) of strawberry *MTA* RNAi plants, but the number of hypermethylated transcripts (617) was lower than that (1239) of strawberry. Because we focused on the hypomethylated genes in *MdMTA* RNAi plants rather than hypermethylated genes that cannot yet be explained, the effect of MTA on m<sup>6</sup>A modification in apple is similar to that in strawberry.

In Arabidopsis, knockdown of *MTA* leads to embryonic lethality, shortened inflorescences, leaf crinkling and reduced apical dominance (Bodi *et al.*, 2012). In rice, *Osmta* mutants show



reduced panicle length, fertility and effective seed number (Zhang *et al.*, 2019). Overexpression of *MTA* in poplar increases the density of trichomes and promotes the development of the root system (Lu *et al.*, 2020). In strawberry, *MTA* regulates nonclimacteric fruit ripening through the abscisic acid (ABA) pathway (Zhou *et al.*, 2021). All these studies shed light on how *MTA* functions in development. In our study, *MdMTA* RNAi plants also had developmental defects, including weaker root systems and shorter plant heights, whereas the *MdMTA* OE plants had comparable root systems and plant height to GL-3 plants (Fig. 5a). The level of *MdMTA* protein in OE plants is only 2.2- to 2.4-fold that of the wild-type plants, which might be not high enough to trigger the developmental process of the transgenic plants (Fig. S8c). However, *MdMTA* OE plants were more tolerant to drought stress than GL-3 plants, indicating that plants may need less *MdMTA* protein in response to abiotic stress than for growth and development. However, this hypothesis needs further exploration.

m<sup>6</sup>A affects multiple aspects of mRNA metabolism, but its most prevalent effect is on mRNA stability (Shao *et al.*, 2021). The relationship between m<sup>6</sup>A and mRNA stability is highly complex. FIP37-mediated m<sup>6</sup>A modification was negatively correlated with mRNA stability (Shen *et al.*, 2016), and VIRILIZER-mediated m<sup>6</sup>A methylation negatively regulated mRNA stability under salt stress (Hu *et al.*, 2021). YTHDF2 destabilizes transcripts that contain m<sup>6</sup>A modification by interacting with the CCR4-NOT complex in the processing body to

shorten the poly (A) tail (X. Wang *et al.*, 2014; Y. Wang *et al.*, 2014; Du *et al.*, 2016). The above results suggest that m<sup>6</sup>A promotes mRNA instability. However, the Arabidopsis ECT2 protein, which is homologous to YTHDF2, can promote m<sup>6</sup>A-mediated mRNA stability (Lockhart, 2018). In the *mta* mutant, m<sup>6</sup>A-modified transcripts are unstable (Anderson *et al.*, 2018). Loss of VIRILIZER-dependent m<sup>6</sup>A inhibits mRNA stability by promoting the selection of downstream proximal cleavage and polyadenylation sites in mammalian cells (Parker *et al.*, 2020). Changes in RNA stability have a significant impact on the regulation of gene expression (Ayupe & Reis, 2017; Shoshani & Cleveland, 2020), indicating that m<sup>6</sup>A can regulate the expression level of methylated transcripts. Our results showed that m<sup>6</sup>A modification could positively or negatively regulate gene expression. m<sup>6</sup>A tended to be positively correlated with upregulation of genes in GL-3 under drought conditions (Fig. 2c-e). m<sup>6</sup>A also tended to correlate positively with gene expression levels in *MdMTA* RNAi plants under normal conditions (Fig. 6a). Interestingly, under drought treatment, the expression level of hypomethylated transcripts in *MdMTA* RNAi plants tended to be downregulated, in contrast to the situation under normal conditions (Fig. 6a,c,d). Further analysis of mRNA stability indicated that m<sup>6</sup>A-hypomethylated transcripts were indeed less stable in *MdMTA* RNAi plants than in GL-3 plants under drought (Fig. 8a). Together, these results suggest a complex relationship between m<sup>6</sup>A modification and gene expression.



m<sup>6</sup>A regulates translation efficiency by multiple mechanisms. The effect of m<sup>6</sup>A on translation efficiency is related to many factors, including RNA structures (Liu *et al.*, 2015; Roost *et al.*, 2015), cellular environment (Zhou *et al.*, 2015, 2018), the recognized protein (readers) (Shi *et al.*, 2017; Zhou *et al.*, 2018) and localization (Meyer *et al.*, 2015). m<sup>6</sup>A modification in the 5' UTR and start codon facilitates cap-independent translation by binding to eukaryotic eIF3 (Meyer *et al.*, 2015). However, the m<sup>6</sup>A modification in the 3' UTR promotes cap-dependent translation through the cytosolic m<sup>6</sup>A readers YTHDF1 and YTHDF3 (Wang *et al.*, 2015; Shi *et al.*, 2017). m<sup>6</sup>A modification in the CDS positively regulates translation by resolving mRNA secondary structures through the only RNA helicase-containing m<sup>6</sup>A reader, YTHDC2 (Mao *et al.*, 2019). The distinct functions of different readers might contribute to the fact that MdMTA affected the translation efficiency of only four mRNAs with the m<sup>6</sup>A modification under drought (Figs 8b, 10d). Our findings indicate that MdMTA influences translation efficiency at the 5' UTR and CDS regions (Fig. 8b); however, whether it also influences translation efficiency at the 3' UTR requires further investigation.

The regulatory mechanisms of methyltransferases during abiotic stress are very limited. We found that the reduced m<sup>6</sup>A levels in *MdMTA* RNAi plants were associated with lower expression of genes involved in lignin deposition and oxidative stress responses (Fig. 8). As one of the main components of secondary cell walls, lignin is a three-dimensional polymer with aromatic properties that hardens the cell walls and makes them impermeable to water (Besseau *et al.*, 2007). Under drought stress, *MdMTA* RNAi plants contained less lignin than GL-3 plants, whereas OE plants contained more (Fig. 9a); H<sub>2</sub>O<sub>2</sub> levels were higher in *MdMTA* RNAi plants and lower in *MdMTA* OE plants than in GL-3 plants (Fig. 9b,c). Consistently, POD and CAT enzyme activities were lower in *MdMTA* RNAi plants and higher in *MdMTA* OE plants (Fig. 9d–i). Our data indicate that MdMTA may positively regulate drought stress tolerance by promoting lignin deposition and antioxidant enzyme activity. We also noted that with extension of the growth period, *MdMTA* RNAi plants had lower lignin content than GL-3 plants under both control and drought conditions, consistent with the shorter height of *MdMTA* RNAi plants (Figs S20a, S22).

In summary, we have identified an RNA methyltransferase and demonstrated the molecular role of m<sup>6</sup>A modification in drought response. We found that MdMTA-mediated methylation positively regulated the expression of genes involved in lignin deposition and ROS scavenging in response to drought stress. Our research provides essential insights into m<sup>6</sup>A modification and MdMTA regulation in drought tolerance (Fig. 11).

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## Author contributions

QG and NH designed the project. NH, CL and SY performed the experiments. JH, YL, LX, FM and MM analyzed the data. QG, MMT and NH wrote the manuscript.

## ORCID

Qingmei Guan  <https://orcid.org/0000-0002-3153-3147>  
Jieqiang He  <https://orcid.org/0000-0002-2617-3159>  
Nan Hou  <https://orcid.org/0000-0001-5860-4470>  
Chaoshuo Li  <https://orcid.org/0000-0002-1643-9048>  
Yu Liu  <https://orcid.org/0000-0003-3371-0087>  
Fengwang Ma  <https://orcid.org/0000-0003-0608-2521>  
Mickael Malnoy  <https://orcid.org/0000-0002-5158-534X>  
Muhammad Mobeen Tahir  <https://orcid.org/0000-0001-7511-0608>  
Lingfei Xu  <https://orcid.org/0000-0003-0200-2811>  
Sisi Yu  <https://orcid.org/0000-0002-5011-5418>

## Data availability

The RNA-seq and m<sup>6</sup>A-seq data have been deposited at the NCBI with the dataset identifier PRJNA772664.

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## Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

**Fig. S1** Pearson correlation analysis of input and immunoprecipitation (IP) reads from the m<sup>6</sup>A peak regions identified by m<sup>6</sup>A-seq in GL-3 plants under drought and normal conditions.

**Fig. S2** m<sup>6</sup>A levels of six transcripts in GL-3 plants before and after drought treatment.

**Fig. S3** Proportion of different m<sup>6</sup>A sequence motifs.

**Fig. S4** Gene ontology (GO) enrichment of hypermethylated and hypomethylated transcripts after drought treatment in GL-3 plants.

**Fig. S5** Pearson correlation analysis of transcripts identified from RNA-seq in GL-3 plants under drought and normal conditions.

**Fig. S6** Protein sequence alignment of MdMTB and AtMTB.

**Fig. S7** MdMTA and MdMTB protein expression and purification.

**Fig. S8** Identification of transgenic plants and detection of m<sup>6</sup>A levels.

**Fig. S9** Pearson correlation analysis of input and immunoprecipitation (IP) reads of the m<sup>6</sup>A peak regions identified from m<sup>6</sup>A-seq in MdMTA RNAi plants under normal conditions.

**Fig. S10** Distribution of m<sup>6</sup>A in MdMTA RNAi plants under normal conditions.

**Fig. S11** Gene ontology (GO) enrichment of m<sup>6</sup>A peaks installed by MdMTA under normal conditions.

**Fig. S12** Pearson correlation analysis of transcripts identified from RNA-seq in MdMTA RNAi plants under normal conditions.

**Fig. S13** Morphology of MdMTA transgenic plants after drought treatment.

**Fig. S14** Photosynthetic indexes of GL-3 and MdMTA transgenic plants in different periods of drought treatment.

**Fig. S15** Pearson correlation analysis of input and immunoprecipitation (IP) reads of the m<sup>6</sup>A peak regions identified by m<sup>6</sup>A-seq in MdMTA RNAi plants under drought conditions.

**Fig. S16** Pearson correlation analysis of transcripts identified from RNA-seq in MdMTA RNAi plants under drought conditions.

**Fig. S17** Distribution of m<sup>6</sup>A in MdMTA RNAi plants under drought conditions.

**Fig. S18** MdMTA-mediated m<sup>6</sup>A enrichment of transcripts under normal conditions.

**Fig. S19** mRNA stability and translation efficiency under normal conditions.



**Fig. S20** Lignin content in *MdMTA* RNAi transgenic plants.

**Fig. S21** MdMTA-mediated m<sup>6</sup>A modification had no influence on the mRNA stability and translation efficiency of CAT and POD enzymes under normal conditions.

**Fig. S22** Morphology of *MdMTA* RNAi plants under long-term drought stress.

**Method S1** Supplementary materials and methods.

**Table S1** Primers used in this study.

**Table S2** Summary of m<sup>6</sup>A-seq results in GL-3 and *MdMTA* RNAi plants under normal and drought conditions.

**Table S3** Different peaks in GL-3 plants under drought conditions compared to normal conditions.

**Table S4** Differentially expressed genes in GL-3 plants under drought conditions compared to normal conditions.

**Table S5** Different peaks in *MdMTA* RNAi plants compared to GL-3 plants under normal conditions.

**Table S6** Differentially expressed genes in *MdMTA* RNAi plants compared to GL-3 plants under normal conditions.

**Table S7** Differentially genes with m<sup>6</sup>A modification in *MdMTA* RNAi plants compared to GL-3 plants under normal conditions.

**Table S8** Differentially expressed genes with hypomethylated m<sup>6</sup>A peaks in *MdMTA* RNAi plants compared to GL-3 plants under drought conditions.

**Table S9** Alternative splicing events in *MdMTA* RNAi plants compared to GL-3 plants under drought conditions.

**Table S10** Differential alternative splicing events in *MdMTA* RNAi plants compared to GL-3 plants under drought conditions.

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